LVMH MuleSoft

LVMH TECH HACKATHON

Theme: Integration / MuleSoft

Agent-Ready Order Management with MCPs

Your mission: make Order Management **agent-ready** — exposing APIs and data as MCPs that an AI agent can orchestrate in natural language.

Date	Location	Format	Tools
April 9, 2026	Station F, Paris	4-hour hackathon	Anypoint Platform · MuleSoft Vibes · Claude Code

Prerequisites

REQUIREMENT	DETAILS
Anypoint Org access	Credentials will be provided — make sure you can log in. Always use **LVMH AI Hackathon** business group (not the root org)
VS Code	Installed and up to date
Anypoint Code Builder Extension Pack	Installed in VS Code
Claude account	Required for Claude Code
Claude Code Extension	Installed in VS Code
Claude Desktop (or similar MCP client)	To test and demo your deployed MCPs

The Challenge

Two APIs. One Database. Agent-Ready via MCPs.

You will be provided with backend systems representing a simplified e-commerce backend typical of LVMH Maisons. Your job is to make them **agent-ready** by exposing them as MCPs that an AI agent can query in natural language.

You are split into two teams. Use `team-1` or `team-2` as a prefix in all your naming (bridges, apps, etc.).

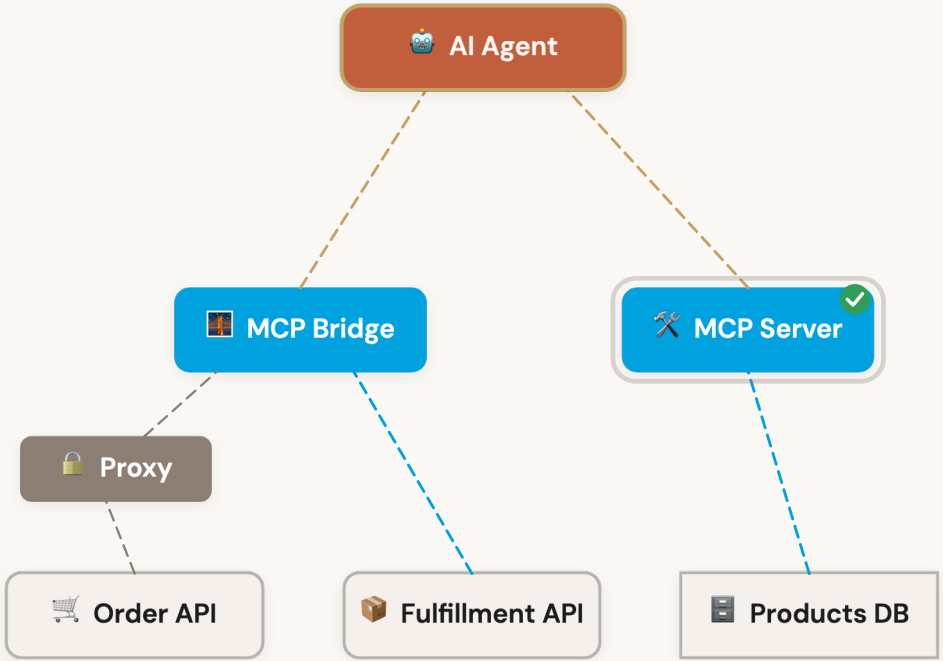
SYSTEM	TYPE	WHAT IT EXPOSES
🛒 **Order API**	🌐 REST API	Orders placed by customers — order ID, customer info, items, timestamps
📦 **Fulfillment API**	🌐 REST API	Shipment and delivery status — tracking number, carrier, estimated delivery
🗄️ **Product Catalog**	🛢️ DB	Product details — name, description, category, price, images

Suggested Build Plan & Architecture

1. **MCP Bridge**
   Bridge the REST APIs via API Manager
2. **MCP Server**
   Vibe-code a Mule app for the Database
3. **Deploy**
   Ship to Anypoint Platform
4. **Test**
   Agent queries all three systems

A. **Data Masking**
   Apply policies on the Order API

B. **Best Practices**
   Refine code to LVMH standards



🌉 Step 1 — Create the MCP Bridge

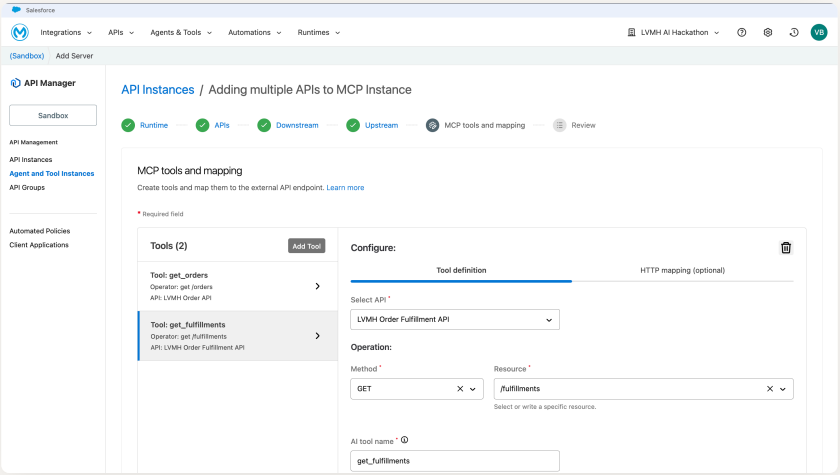
In API Manager, create an MCP Bridge on top of the **Order API** and **Fulfillment API**.

Browse each API spec and **select the endpoints to expose as tools** — you can select all of them.

The bridge translates your selection into MCP-compatible tools — no code required.

Deploy it on the **Ingress Flex Gateway** provided in your environment.

📖 MCP Bridge docs



Selecting endpoints in the MCP Bridge UI

🛠️ Step 2 — Vibe-Code the MCP Server

Using **Claude Code** with the ACB extension, vibe-code a Mule application that connects to the **Products Database** and exposes it as an MCP server.

Use these details to connect to the existing MySQL **Products Database**:

🔐 Connection

```
host: cloud-services.demos.mulesoft.com
port: 31717
username: root
password: Autom-tionR0cks.
database: lvmh
```

🛢️ Table `product_catalog`

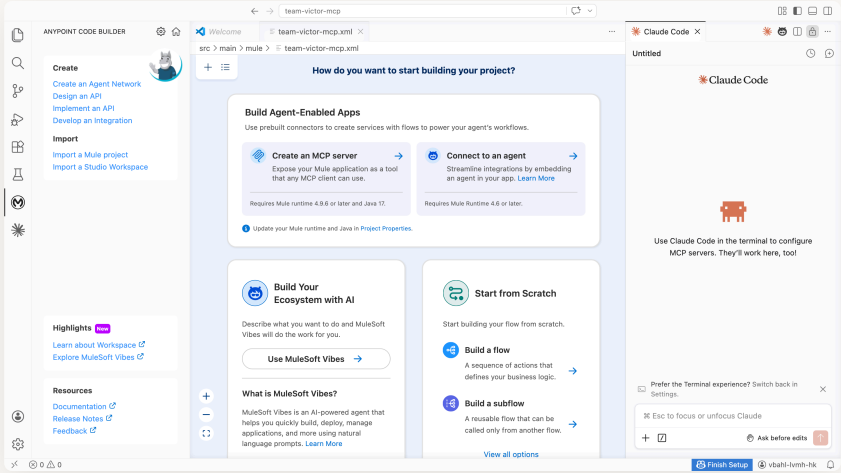
```
id            INT AUTO_INCREMENT PRIMARY KEY
product_id    VARCHAR(50)    -- PRD-XXXXXXXX
sku           VARCHAR(50)
name          VARCHAR(255)
brand         VARCHAR(100)
category      VARCHAR(100)
subcategory   VARCHAR(100)
price         DECIMAL(10,2)
currency      VARCHAR(3)     -- default EUR
stock_status  ENUM('IN_STOCK','OUT_OF_STOCK','LOW_STOCK')
created_at    TIMESTAMP
```

🛠️ Step 2 — Vibe-Code the MCP Server

Try different prompt strategies — iterate and refine.

Experiment with what works best.

> *Claude is great at general Mule code, but newer components (like MCP) may not be in its training data.*
>
> *Hint: You could ask it to **search the web for examples** before generating unfamiliar syntax.*



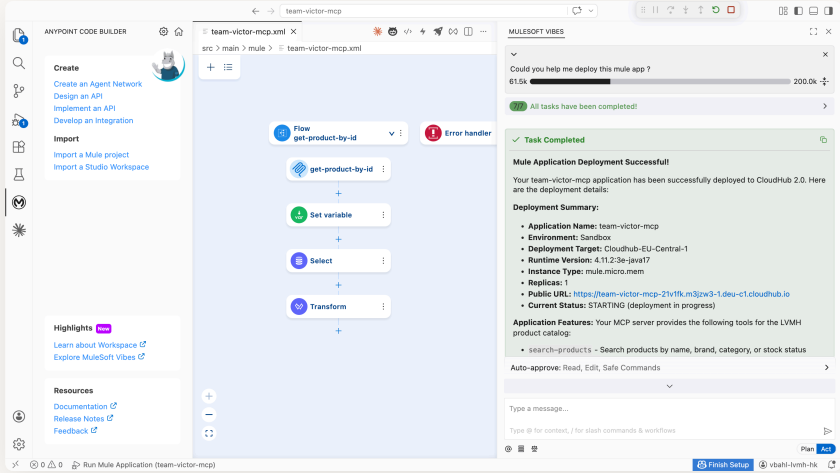
Claude Code in VS Code with MuleSoft Extension Pack

🚀 Step 3 — Deploy to Anypoint Platform

Deploy your MCP server to the provided Anypoint environment using **MuleSoft Vibes**.

Ask Vibes to deploy your application — it handles packaging, environment selection, and deployment through natural language commands.

MCP Server must be live and callable.



MuleSoft Vibes in VS Code

🧪 Step 4 — Test with an AI Agent

Add both MCPs to **Claude Desktop** via the config file (`claude_desktop_config.json`), *not the UI — the bridge URL may not end with* `/mcp` .

```
{
  "mcpServers": {
    "mcp-bridge": {
      "command": "npx",
      "args": ["mcp-remote", "<your-bridge-url>"]
    },
    "mcp-server": {
      "command": "npx",
      "args": ["mcp-remote", "<your-server-url>"]
    }
  }
}
```

Test a live conversation that queries **all three systems** in a single interaction.

Try this prompt:

```
Show me the 3 latest orders with shipment info,
and the full product details for every item ordered.
```

🔒 Bonus A — Data Masking

⚠️ Your agent sees **everything** — customer emails, addresses, phone numbers…

In production, that could be a problem.

```
{
  "name": "T** C***",
  "email": "t*****@g****.com",
  "phone": "+33 * ** ** ** **",
}
```

> 💡 *Start with just **one field** — you can always add more later.*

Suggested steps:

1. Create an **API proxy** of the Order API in API Manager. Use the **label** to identify your team.
2. Apply a **data masking policy** on the proxy using MuleSoft Vibes
3. Update your **MCP Bridge routing** to point to the proxy instead of the original API
4. Test that sensitive field is masked in the agent's response

📏 Bonus B — LVMH Best Practices

On Anypoint Exchange, you will find an MCP server connected to Confluence that exposes some LVMH's integration standards and best practices.

Add this MCP to your environment and re-generate or refine your MuleSoft MCP Server to follow LVMH standards.

Compare the before and after.

Sneak peek:

- `LOG_001` Log Before External Call
- `LOG_002` ...
- `LOG_003` ...
- `ERR_001` Wrap External Calls
- `ERR_002` ...
- `ERR_003` ...

> *Connect the MCP to find out more.*

🎤 Show & Tell

Prepare a short presentation with a **live demo** of your agent conversation end to end.

Tell us:

- **Live demo** — show your agent querying all three systems in one conversation, your IDE / flows
- **The journey** — your vibe-coding process, prompts that worked (and didn't), how you iterated
- **What surprised you** — moments that changed how you think about prompting, building integrations…
- **What's next** — if you had another 4 hours, what would you add?

Be fun. Be creative. 🎉

> ***Need a slide deck?*** *Fork this one — built with Slidev and mainly vibe-coded with Claude Code* 😊